

# Democracy and Alternatives: Exploring Political Systems through Agent Based Modelling

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## Abstract

The main object of this paper is to investigate the implications of changing the political system from democracy to epistocracy. In the paper, democracy as well as several alternatives are defined, where epistocracy is deemed the best alternative in two forms. To investigate the implications, two separate Agent Based Models are carried out to simulate first the gathering of information, informing the vote of sovereign agents, where after the voting itself is carried out. Before the ABMs are constructed and evaluated, a comprehensive justification for trying another political system is presented along with a presentation and definition of the alternatives. Two hypotheses are formulated on this foundation, both hold that the two different forms of epistocracy will result in better governance than democracy. A data analysis is conducted on the data from the ABMs by inspecting the data visually as well as regression models. The findings imply no significant better outcome of changing the political system, although a more consistent better outcome can be argued. Caution is warranted when interpreting these results, due to the limitations associated with agent-based modelling when investigating social dynamics.

**Keywords:** Political Theory, Democracy, Agent Based Modelling, Epistocracy, Governance

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# 1. Introduction

In current times, with a world where the political and economic center of gravity is drifting towards the east, the geopolitical world order is becoming increasingly multipolar (Quah, 2011). Each country will find itself with a choice of which political system they best see fit. As we shift away from the dominance of the western world, with the EU falling behind in economic growth, needing serious investments to keep pace with China and the USA (Blenkinsop, 2024; Draghi, 2024), this is even more imperative. In the previous two centuries, democracy has been the most common answer that most nations gravitated towards after various amounts of trial and error. This is a result, according to historians and political scientists, of three separate waves of democratization happening since the 18th century. These waves have combined shaped and solidified democracy as the default political system. The first happening in the 19th century, described as the slow wave, after a brief resort to authoritarianism, the second wave sat in just after the second world war and the third beginning in the early 1970's until early 1990's (Doorenspleet, 2000; Huntington, 1991).

In recent years however, we have seen increasing anti-democratic trends across the world. In 2024, a former president of the United States of America was convicted on 34 felonies of fraud and defamation. Additionally, he was charged with election fraud and subversion of the 2020 election, which culminated in the attack on the capitol on January 6th 2021 (Donald J. Trump [@realDonaldTrump], 2020). The current ruling on the subversion of the election from the United States Supreme Court, has that a former president should be immune to prosecution for official acts (Graham, 2024).

Further examples include the Indian people's loss of civil liberties and political rights following Narendra Modi becoming prime minister in 2014. The Hindu nationalist government has exerted pressure on human rights organizations under his leadership, as well as facilitating a hostile environment for academics, journalists and muslims, seeing a rise in attacks rooted in bigotry (Serhan, 2022). The story continues with Viktor Orban, prime minister of Hungary, abusing his power during the Covid-19 pandemic to withdraw funding of municipalities governed by the opposition as well as blocking them from getting a fair hearing in the media during the 2018 election and winning again in 2022 (Scheppele, 2022; Wallenfeldt, 2024). Most recently, the president of South Korea, Yoon Suk Yeol, declared martial law in an attempt to consolidate power (Klug, 2024).

All these happen more or less within the democratic framework, where clever people find ways to manipulate and utilize the voters to bring forth their anti-democratic visions.

It is not a new phenomenon either. During the 20th century, this happened on numerous occasions as well, in the interim between the democracy waves defined by Huntington in 1991, called the first and second reverse waves (Huntington, 1991). The waves happened in one of two ways mostly; revolution, like Mussolini's Italy, where he seized power by first being elected into the lower chamber

of parliament and then marching on Rome and forcing the King to allow him to form a new government with him as the supreme leader (Editors, n.d.-b; Frommer, 2023). The second way, is the one most like today's clever minds use, pioneered by Hitler, when his revolution coup failed. He got himself elected to government and slowly edged out everyone else until he was elected chancellor and declared a state of emergency making himself the supreme leader and could turn Germany into a fascist authoritarian regime (Editors, n.d.-a; ww2museum, n.d.). Even Lenin somewhat used the framework of democracy to gain power, when he deliberated for 10 hours about starting a violent revolution in Russia and won the vote 10-8 (Bang et al., 2024). Effectively, Lenin used both strategies. The cause for these kinds of antidemocratic trends within the democratic frame is manifold. One such cause is that the general electorate is wildly un- or misinformed. Most voters know nothing or less than nothing. In his book "Against Democracy", Jason Brennan outlines the systematic way in which voters are uninformed and misinformed. He highlights the example of voters in the UK when they were voting on whether the UK should leave the European Union in 2016. In this election, "Leave" won by a small margin. Brennan references a poll conducted by Ipsos Mori a month before the decision. Here voters were systematically misinformed in regards to facts relating to the decision that lay before them. Both Leave voters and Remain voters overestimated how large of a percentage EU immigrants comprised of the population, Leave averaging a guess of 20%, where Remain averaged a guess of 10%, the real number was closer to 5%. Both were also off by a factor of 40 to 100, on average about how much the UK paid in Child Benefits to other countries. Both also vastly underestimated the foreign investment from the EU and overestimated the foreign investment from China (J. Brennan, 2017; Ipsos, 2016). Recent debates also highlight this. The presidential debate between Kamala Harris and Donald Trump, where both candidates outright lied on numerous occasions, which also warranted the largest amount of fact checking journalists ever, 27 (Dalmia, 2024; Gilder et al., 2024; Waldman, 2024). Other examples of the voters actually being misinformed on both sides of the aisle in the USA are the recent "Surrounded" debates from Jubilee, where a conservative debates 20 liberals in one and a liberal debates 20 conservatives in another. In both debates, debaters were presenting arguments based on misinformation or even making stuff up on the spot. This was also fact checked for the videos, with fact check info boxes showing up everytime a false claim was uttered (Jubilee, 2024a, 2024b).

This begs the question: Is democracy still worth pursuing?

In this paper I will investigate the pros and cons of democracy, I will define alternatives and compare them based on a set of chosen criteria. For this purpose I will utilize Agent-Based Modelling (ABM), to simulate an electorate and the political system in both democracy and the defined alternatives. With this, I aim to provide a clearer picture of the state of global politics and potentially start a discussion on whether our current model of governance needs to be reevaluated. I analyze and discuss the outcomes of these ABMs to ensure a well-informed and unbiased basis for readers to think about the governance of their own state.

## 2. Defining democracy and alternatives

To evaluate whether democracy is the best choice, we must first define what democracy is along with possible alternatives.

### 2.1 Defining democracy

The matter of defining democracy is a complex one. There is not necessarily a universally agreed upon definition. For the purposes of this paper, the definition of democracy will be the prevailing form of democracy around the world. The representative democracy. This is formed in a system where the election process allows the citizens to vote for politicians who will represent them in a free and fair election. This means that the execution of the election process should not be hindered in any way by either party. More specifically, the opposition should not be hindered by the incumbent, which is the most likely scenario for hindrance. Moreover, a democracy will have equal opportunity for all, protection of rights, and equal application of the law. This definition is put together from several sources including democracy research organization, Freedom House and two books by political theorist and philosopher, Jason Brennan (J. Brennan, 2017, 2023; Freedom House, n.d.).

This definition excludes nations like Russia and North Korea who claim to be democracies, because democracy is seen as the inherently “good” system (J. Brennan, 2017). So nations like Russia and North Korea claim this for legitimacy. The reality however is that Freedom House defines both as authoritarian regimes with power consolidated around one individual (*North Korea*, n.d.; *Russia*, n.d.).

### 2.2 Defining alternatives

When defining alternatives, they will need to be justified as viable alternatives to democracy, without compromising basic human rights. The alternatives can be defined as belonging to one of two groups, the first group consists of systems of government that have previously been tried in history, such as oligarchy, absolute monarchy and autocracy. The second grouping consists of not yet tried systems of government, this includes various forms of epistemic democracy, also known as epistocracy.

#### 2.2.1 Previously Tried Systems

##### 2.2.1.1 Oligarchy

Oligarchy is defined as the rule of the few. This is when a small group of usually elite people hold all the power. It is generally an authoritarian form of government, where few wealthy people exercise the power for their own good and as a means to accomplish their own ends, seeking to climb to the top of the oligarchy and sometimes in corrupt ways. Usually the way into the ruling class in an oligarchy is

hereditary (15.3B, 2018; Merriam-Webster, 2024; Young, 2024). Notable examples of oligarchies include the Roman Republic (509 BC - 27 BC), although characterized as an early form of democracy, it was a de facto oligarchy with a small group of senators vying to scale the ladder of power and glory. New people in the senate was such an unusual event that it had a name; Novus Homo (Lex Fridman, 2024; *Roman Republic*, n.d.).

#### 2.2.1.2 Autocracy and Absolute Monarchy

Autocracy and absolute monarchy can be grouped in the same broad definition as absolute monarchy is an autocratic system of government. Absolute monarchy is when a royal monarch holds absolute power and ultimate authority. An autocracy is when a single person holds all power and authority. The difference is that the absolute monarchy is usually ruled by a divinely chosen monarch, meaning that the ruler has a heavenly mandate, according to themselves, where the autocracy only needs a person to consolidate their power for whatever reason (*Autocracy*, n.d.; Perry et al., 2023). Examples include the Roman Empire, North Korea, almost every nation in Europe in the middle ages. This form of government was used for a long time and was the default system throughout most of human history.

### 2.2.2 New Approaches

#### 2.2.2.1 Value-only Voting

This system of government is a midway point between democracy and epistocracy. In the value-only system, citizens will be tasked with determining the aims of society, where legislators are tasked with devising the means and implementing policies to achieve this. This functions as a type of democracy, where the voting power is equally distributed (Christiano, 2008).

#### 2.2.2.2 Restricted suffrage

In a restricted suffrage epistocracy, the basis is that one must earn the right to vote. This means that no one has the right to vote by default. Citizens will enjoy extensive civil liberties rivaling the best democracies of the real world, to exercise political speech, to protest, to publish political ideas, but not vote. To earn the right to vote, one must demonstrate a basic level of political knowledge. This is based on the way driver's licenses, medical licenses and so on are acquired. One must pass a test, as it affects other people if one is given the privilege without having the ability (J. Brennan, 2017).

#### 2.2.2.3 Plural voting

In *Considerations of Representative Government*, John Stuart Mill proposes a form of government with a plural voting system. In this, a citizen gets one vote by default and by performing certain actions, they can acquire additional votes. Mill proposed distributing additional votes to people with

certain academic degrees. This could take forms of giving everyone a vote at age 16 and then giving them 5 more upon completing a highschool level education, 5 more after obtaining a bachelor's degree and 5 more after a master's degree. Or we could give everyone a vote at the age of 16 and then 10 more after passing a voter qualification exam (J. Brennan, 2017; Mill, 1861).

## 3. Comparing Political Systems

### 3.1 Comparing Alternatives

In this section the different alternative political systems will be put against each other in order to determine which alternative to democracy is most viable.

First let us take a look at the already tried systems. All three, oligarchy, autocracy and absolute monarchy suffer from the same fatal flaw that deems them dead on arrival, compared to democracy. The power is too concentrated. It doesn't allow for enough diversity to be beneficial, corruption can too easily take hold and the general populace does not actually have anything to say in the way their country is run. As the Hong-Page theorem has it, diversity in leadership produces innovative policies and prevents concentration of power and sometimes two less smart heads are better than one smart head (J. Brennan, 2017; Hong & Page, 2004; Kollman, 2003; Page, 2010). When a large and diverse group seeks a solution to a problem, it will often lead to better decisions than a smaller more expert group, because it can pool the collective diverse knowledge, which in the aggregate is greater than the smaller group (J. Brennan, 2017; Hong & Page, 2004; Somin, 2016).

Does this mean that more diversity is always better? No, not if you ask Scott Page himself. Too much diversity can lead to chaos, inefficiency and even conflict and collapse (Page, 2010). Researchers in multiple disciplines like Jason Brennan, Ilya Somin and Abigail Thompson are also skeptical about too much diversity and doubt whether the Hong-Page Theorem actually justifies democracy as it can sometimes be used for (J. Brennan, 2017; Landemore, 2017; Somin, 2016; Thompson, 2014).

This leads to the not yet tried systems. Value-only voting is in reality a type of democracy and can therefore be ruled out as a *de facto* alternative democracy. If this is not enough however, we might look at the workings of it. A criticism of democracy is that democracy is not competent. If we were to grant that for the sake of looking at value-only voting as an alternative, value-only voting doesn't actually alleviate that problem. If the criticism is that the general electorate is not competent and knowledgeable in the democracy of the present, then they cannot determine what constitutes good means to their ends. This is the problem that Christiano tries to solve by splitting the division of political labor; it sort of works like a yacht owner hiring a captain to sail his boat and telling the captain where to go. The difference is that if an electorate is not competent enough to decide the means of achieving their ends, they will not be competent enough to notice if the politicians don't



steer the country in the desired direction. At least the yacht owner can tell whether the captain has gotten them to their destination or they end up in the wrong place.

The final two systems, restricted suffrage and plural voting hold up against the competency criticism and the diversity criticism. Both base their system of government on the electorate being competent enough, by setting expectations and requirements for the right to vote. Both are diverse enough by including the vast majority of citizens - depending on how high the bar is - and not discriminating on the basis of unchangeable factors like gender and race. Everyone can achieve the level of knowledge they need to require a “voting-license”, just like everyone can achieve the skills and knowledge needed to acquire a drivers license.

This now leaves three systems of government to consider; democracy, restricted suffrage epistocracy and plural voting epistocracy. I will compare these to each other next and justify epistocracy as a viable alternative to democracy.

## 3.2 Democracy vs Epistocracy

With determining both plural-voting and restricted suffrage initially as being the most viable alternatives, they must be compared to democracy to make sure they can indeed be alternatives. For the purposes of this section, epistocracy will be a covering term for both of these political systems and will be compared to democracy in the form it is described in section 2.1.

To further justify epistocracy as a viable alternative I will present the following tenets, posed by David Estlund and Jason Brennan:

- *Truth Tenet*: There are correct answers to (at least some) political questions.
- *Knowledge tenet*: Some citizens know more of these truths or are more reliable in determining these truths than others.
- *Antiauthority tenet*: When some citizens are morally unreasonable, ignorant or incompetent about politics, this justifies not permitting them to exercise political authority over others. It justifies either forbidding them from holding power or reducing the power they have in order to protect innocent people from their incompetence. (J. Brennan, 2017; Estlund, 2008)

To compare them, we must first define a success criteria. For the purposes of this paper the success criteria will be whether a political system complies with the 30 articles of the Universal Human Right Declaration (Nations, 1948), with the exception of article 21 section 3, which states that the election should be by universal and equal suffrage. This makes this section of the article completely and inherently tied to democracy as any other system will violate this. I will argue why section 3 of article 21 shall be excluded further down.

As democracy is the current system and is, as argued in section 2.1, seen as an inherent good system of government and something to strive for, I will tackle the three best arguments for democracy.

There are 3 ways in which democracy can be considered valuable (J. Brennan, 2017):

1. *Epistemic*: Democracy and widespread political participation are good because they lead to just, efficient and/or stable outcomes.
2. *Aretaic*: Democracy and widespread political participation is good because they educate, enlighten and ennoble citizens
3. *Intrinsic*: Democracy and widespread political participation are goods as ends in themselves.

I will tackle these before moving on to epistocracy. First claim is that democracy is epistemically good. It leads to just, efficient or stable outcomes. This is true in many cases. Democracy, at least as it is defined in this article, ensures free and fair elections, where every citizen of age is part of the system of government in their function as an electorate that elects, hopefully, competent leaders to steer the country in the right direction.

However, in many cases the electorate elects a governing administrative body that does not comply with the human rights declaration or even just does not carry out what they were elected for. This has happened multiple times across history, as described in the introduction, where a candidate is elected within the confines of democracy and uses this to subvert the political system, resulting in “undemocratic” governing forms. This can happen both as a result of incompetence and deliberate exploitation for power grabbing (J. Brennan, 2017; Sharma, 2018; Tam, 2019).

The second claim is democracy as an aretaic good. Political participation enlightens and educates citizens. This mirrors J.S. Mill’s education argument (Mill, 1975). If this was true, however, all voters would be informed and know what they were voting on. They would understand the political systems and be able to filter out misinformation. The reality, however, is that most voters are either ignorant or misinformed (J. Brennan, 2017). It is simply not worth it for most voters to be informed. Not much happens if a voter indulges in a conspiracy theory or makes an honest mistake. Any single voter, statistically, does not make a difference. As with planes, any single rivet on the plane is not important, but if it loses too many rivets, it crashes. This is also the case for a system of government. Any single voter in the electorate does not make a difference, but if most voters are excluded, it will have an effect on the governance (J. Brennan, 2017; Page, 2010). It is simply not cost-benefit efficient to be informed. It takes time and effort to stay informed, which most voters simply will not allocate (J. Brennan, 2017). Several studies have found that political participation does not educate nor ennoble voters, some even found it corrupts voters (Birch, 2008; Lever, 2008, 2010).

The third claim is that democracy is inherently and intrinsically good. This claim is on its surface even morally objectionable. As most voters know nothing or even less than nothing and are not incentivized to change this, the universal, unrestricted suffrage results in a system of government, where the voters make decisions in irrational and ignorant ways and then impose this on innocent people. The diversity in information acquisition collapses the good governance of the democracy. As is seen multiple times throughout history and the world (J. Brennan, 2017; Page, 2010).

With this said, democracy, for all its flaws, is still the best performing system we have had so far. It has assured free and fair elections, for the most part, and it gives the people a feeling that they are part of the governing process.

Now moving on to epistocracy. As epistocracy is the “challenger”, I will evaluate the best arguments against epistocracy.

The critiques of epistocracy as an alternative to democracy include:

1. If an epistocracy is implemented, it will be the rule of the elite, as the less informed people will not be heard, this will lead to a system, where the elite rules and the least fortunate are suppressed (J. Brennan, 2017).
2. Political knowledge is not shared equally by demographic groups and therefore certain demographics will be disadvantaged in an epistocracy (Carpini & Keeter, 1996).
3. People will be affected by policies they have no say in. The universal suffrage principle is violated, as it will not result in “one person, one vote” (J. Brennan, 2017).
4. Even if individuals seek or are selected for better performance, there is no guarantee for better outcomes (Hardin, 1968; Page, 2010).

I will now engage with these. The first two objections revolve around the exclusion of certain demographics, including african-americans and poor people. It posits that the exclusion of the less fortunate and certain demographics will suppress these people, as their voices will not be heard, when they don’t have a vote. The electorate in an epistocracy will be whiter, better employed, richer and more male than the electorate of a democracy. This is true, but it is not an actual objection to epistocracy.

As demonstrated in *Against Democracy*, it matters how you vote, not that you vote. The example given is that a young african-american woman posits the second objection. Jason Brennan asks her if he were to wave a magic wand that would make all black people vote for Trump and republicans and in any future elections, they will vote for the equivalent of Trump or candidates that supported policies she didn’t like, would she want him to do it. The answer was no. When members of a group vote, if they are misinformed, they might not even help themselves, but actually disadvantage themselves. An example is that Donald Trump’s policies are not actually good for his main demographic, white rural Americans (J. Brennan, 2017; Downs, 2024; Willingham, 2020).

So how do people vote? Several studies on how people vote conclude that people vote for what they perceive to be the common good, rather than their own self-interest (G. Brennan & Lomasky, 1993; J. Brennan, 2017; Funk, 2000; Funk & García-Monet, 1997). So if the white, rich, male electorate is more informed as to what constitutes the common good, epistocracy will actually be better for the less fortunate.

This leads to the third critique of epistocracy, which is also where I will argue for the exclusion of article 21, section 3 of the Human Rights Declaration. This critique focuses on the removal of universal suffrage, resulting in people having no “say” in the way their country is governed. This

objection is at best a moral objection. As it stands now, people have no actual “say” in the way a country is governed. Unless you are a politician, you have no real influence at all. The odds of a single vote determining an election result are astronomically small. Even in a small country like Denmark that has just over 4 million voters (Boeskov & Selmer, 2022) and even if the election was decided between two parties with approx. 50% of the votes each, the odds would be 1 in 3.7 million, assuming average participation for danish parliament elections of about 86% (Olsen & Larsen, 2022). The odds are even smaller in huge countries like the USA. So given that voters vote for the perceived common good and that any individual vote makes no real difference, universal suffrage makes no sense. In reality the objection becomes more of an “I feel like I should have this, because I am used to it” kind of objection. This is why article 21, section 3 can be excluded as it inherently ties the universal declaration of Human rights to democracy without just cause. If anything the voting rights of ill-informed voters contributes to this exact hostage-holding behavior that the critique objects to. When ill-informed voters vote for less qualified candidates with bad policies, it holds the rest hostage and forces them to accept the outcome that would most likely have been avoided if the ill-informed and ignorant didn’t vote. It becomes a tyranny of the ignorant masses. This leads to the final objection against epistocracy. This objection holds that even if individuals seek or are selected for better performance, there is no guarantee for better outcome. This is perhaps the best objection to epistocracy. Why change something that is clearly working most of the time? Something that is the best we have had so far. The answer is, because it has been proven that informed voters make better decisions, on multiple occasions (Banerjee et al., 2012; Blais et al., 2009; Pande, 2011; Sanders et al., 2021; Sinclair & Plott, 2012).

### 3.3 Hypotheses

The following hypotheses for changing the political system are posited:

**H1:** Changing the political system to a restricted suffrage epistocracy, ie. excluding voters based on information-level, will result in a better outcome than democracy.

**H2:** Changing the political system to a plural voting epistocracy, ie. giving certain voters extra votes, based on information-level, will result in a better outcome than democracy.

## 4. ABM Theory

To truly compare the political systems I have utilized Agent Based Modelling (ABM). This was chosen as ABMs provide a way to explore dynamic and complex systems in a versatile way, thus they allow for simulation of non-linear interaction between autonomous entities with individual decision-making capabilities and behaviors, known as agents (Bonabeau, 2002; Elsenbroich, 2011; Macal & North, 2010). In contrast to linear models, which are often used in quantitative research in

the social sciences, when utilized well, ABMs can offer valuable insight into the complex mechanics of interactions between agents. These are encoded with theoretically justified inclinations, with inclusion of the characteristics the environment that the agents operate in provide (Alicea et al., 2023; Rocca & Tylan, 2022). Linear models offer little complexity and explanatory power for non-linear phenomena, regardless of the perceived complexity of the model, which are attributed to the number of parameters included in the linear model (Mehdizadeh et al., 2022; Thorve et al., 2022). Therefore, using ABMs has the advantage of giving a bottom-up approach that can capture the micro-level behavior of individual agents which then in turn forecast the macro-level behavior of people. As such, this effectively tests whether the theories and mechanisms used to explain the behavior, can provide sufficient measures to understand the empirical evidence (Helbing & Balmelli, 2015; Kiesling et al., 2012; Mehdizadeh et al., 2022).

In relation to democracy and voting, ABMs have on multiple occasions and in multiple countries been used to forecast the turnout of elections, which have been compared to the real election turnouts. Furthermore, they were fitted with behaviors that are encoded from the beginning as well as influences by other agents in the model, thereby introducing a social aspect (Palombi & Toti, 2015; Singh et al., 2011; Yap, 2021). These models have been used with great success and provided a new way of forecasting elections, analyzing voter behavior and, importantly for this paper, simulating interventions in voting eligibility (Fowler & Smirnov, 2005; Gao et al., 2022; Kiesling et al., 2012; Qiu & Phang, 2020). Using ABM for this paper allows for multiple interventions and social interactions in information gathering and voting, thus making the simulation more akin to a realistic election cycle. Moreover, the use of ABM allows for basing the voting and information gathering behavior on multiple parameters. In the interest of making the ABM as useful as possible there needs to be theoretically justified deliberate assumptions about the complex mechanisms being modelled (Helbing & Balmelli, 2015). These will be laid out in the next section.

## 4.1 Information gathering and voting behavior

For this paper, two ABMs were constructed. One for information gathering and one for voting. The section will therefore be divided into theoretical parameter selection for each. The workflow of the ABM and the parameters will be described in the methods section.

### 4.1.1 Information gathering

The information landscape is ever shifting and people try to keep up the best they can. Different people gather information in different ways. Reviewing the literature on information gathering, some patterns emerged. People will generally seek out information according to their prior beliefs (Jordan B Peterson, 2024; Millner et al., 2020; Nickerson, 1998) this can be due to watching the same opinionated news channel every day, like Fox News or CNN or having a tailored algorithm on social

media that only feeds the content you have indicated an interest in, like TikTok (Cervi et al., 2023). Generally, when we as humans seek out information it is to clear up something that we don't know, so when we are in the echo chamber, where our confirmation bias is being nursed with only agreeable information, we don't actively seek out new information and we therefore become complacent and the information landscape moves on without us (Li et al., 2024; Ruppel et al., 2020).

#### 4.1.2 Voting Behavior

Although voting behavior can be quite complex and have differences across age, sex, ethnic groups etc. The real interest for the purposes of this paper is to see if epistocracy will do better than democracy, wherefore the main voting behavior examined is that of informed vs uninformed people. Research suggests that the following holds true of informed voters, why the opposite will hold true of uninformed people (J. Brennan, 2017):

- Less government intervention
- Less government control of economy
- Pro free trade
- Against protectionism
- Pro-choice
- Tax-increase to off-set debt and deficit
- Soft on crime
- Less hawkish on military policy
- Favor other interventions instead of military
- Accepting of affirmative action
- Against prayer in schools
- In favor of market solutions to healthcare
- Less moralistic in law

Additionally, people will put different emphasis on different issues. Some people will put more emphasis on the candidates' stance on education, some will put more emphasis on the economy. This usually comes down to external factors like job, wealth, ethnicity, gender, age etc. (Andersen & Reiermann, 2019; Chris Williamson, 2024; ICPSR, 2004; Reiermann & Andersen, 2019; Rosenkilde, 2022; Ross Arguedas et al., 2022) Further, these also fit the declaration of human rights. Examples are articles 5-11 for being soft on crime, articles 5, 12 and 14 for being less hawkish on military policy, article 18 for being against prayer in school and less moralistic in law and article and articles 3 and 12 for being pro-choice (Brown, 2016; Nations, 1948; WHO, 2023).

## 5. Methods

To investigate the effects of changing the political system, two ABMs were made. The ABMs were built in NetLogo (Wilensky, 1999) and Python (Van Rossum & Drake, 2009), using the mesa module (Kazil et al., 2020), respectively. The code for the ABMs as well as data analysis can be found in the Github repository Bachelor-project (von Essen-Müller, 2024/2024).

### 5.1 Parameters and characteristics for ABM 1

#### 5.1.1 Agent characteristics

The characteristics of an agent are crucial for differentiating between the agents of the model. The choice of characteristic is guided by the aim of using the model to have agents get different information-levels.

##### 5.1.1.1 Information-level

Each agent is assigned an information-level of 1 to simulate equal ignorance at the beginning. This is subject to change with the parameters of the model. The information-level is limited to the range of 0.5 to 100, signaling the ability to know at most everything and the inability to know absolutely nothing or less than nothing. In conjunction with this, each agent is also assigned an information-level group consisting of quartiles from 0-100.

##### 5.1.1.2 Credibility

Each agent is also assigned a credibility of 1-10 randomly for their interactions, to simulate the difference in persuasion, knowledge and credibility that people have in real life and therefore their ability to change other people's minds. This characteristic is static. Furthermore, agents are also assigned a trust value, which is dynamic and changes based on the credibility of the agents they meet.

##### 5.1.1.3 Opinions

Agents are assigned opinions of numeric value of 0-100 based on their information-level group. The opinions are formulated after the opinions of informed voters laid out in section 4.1.2. Further, agents are assigned a weight from 0-1 to each opinion at random, to simulate the perceived importance that different voters put on different issues.

#### 5.1.2 Parameters

Some global parameters were introduced into the model.



#### 5.1.2.1 Number of people

A slider was introduced to alter the number of people in the simulation.

#### 5.1.2.2 Terrain

The “terrain” or the physical aspect of the model is that of an “information landscape” where agents travel and reach both peaks and valleys. This is shuffled randomly in each new simulation to represent different information landscapes in different societies and systems. However, the peaks were set to be every 100 patches and branch out in a gradient of 12 patches in each direction. The effects will be explained in section 5.1.3.

### 5.1.3 Agent interactions

The model also features some agent interactions to observe the impact of other people in information gathering.

#### 5.1.3.1 Persuade

This parameter is the basic interaction for informing other agents. When an agent encounters an agent with a higher or lower information-level than itself, it gets more or less informed accordingly.

#### 5.1.3.2 Trust

The trust parameter determines the influence each agent has on the other agent, based on the credibility of the agent it encounters. If the other agent has a credibility of more than 7, it adds .005 to the trust, if the other agent has a credibility less than 4, it subtracts .003 from the trust. The parameter is combined with the persuade parameter by multiplying the impact in the persuade parameter by trust. This is based on the evidence that people trust their peers more than experts to get their information and that we trust the advice of friends and family more than strangers (Grimmer, 2016).

#### 5.1.3.3 Agent-Patch interactions

There are two agent-patch interactions, the first interaction changes the information-level of the agent when they reach a peak, walk away from a peak and reach a valley. The other is the change-opinion interaction, that adds or subtracts to the numeric value of opinions based on stepping on a peak or in a valley.

#### 5.1.3.4 Impact of information-level group

The agent-agent interaction impact is different from agent-patch interaction. Stepping on a peak, affords + 5 information-level, moving away from peaks gives - 0.001 information-level per step and



entering a valley gives - 0.5 information-level per step. These are signalled by white, green and black respectively in the landscape.

Each information-level group reacts differently to the persuasion from other agents. This can be seen in the function pictured in appendix 1.

## 5.2 Parameters and characteristics for ABM 2

### 5.2.1 Agent characteristics

For the second ABM there are two different types of agents, Voters and Candidates. The Voters have all the same characteristics as in ABM 1, but only information-level and opinions are used for this ABM. The candidates only have opinions that consist of the same numeric opinions as the voters. The candidates were however installed with extreme opinions, candidate 1 holds extremely good opinions and candidate 2 holds extremely bad opinions.

### 5.2.2 Parameters

#### 5.2.2.1 Number of steps

The number of steps needs to be specified, when running the model.

#### 5.2.2.2 Information-level threshold

This parameter determines the threshold of information-level of the agents in the ABM.

### 5.2.3 Agent interactions

#### 5.2.3.1 `evaluate_candidate()`

There is only one agent interaction in this ABM, the `evaluate_candidate()` function. This function uses euclidean distance between the summed numeric opinions of the candidates and the summed weighted opinions of the voters. The function itself can be seen in appendix 2.

## 5.3 Workflow of both ABMs

Agents are imbued with information-level of 1, credibility and opinions. The agent then moves a set distance in any direction. The agent then checks for patchcolor and if there is another agent present. They then update their own information-level accordingly and then move on. Once this has run for the set number of ticks. The data is extracted and placed into the voting ABM, where agents immediately encounter a candidate and then evaluate the euclidean distance between their own weighted opinions

and the candidates opinions. If the euclidean distance is smaller than the candidate before, they choose this candidate.

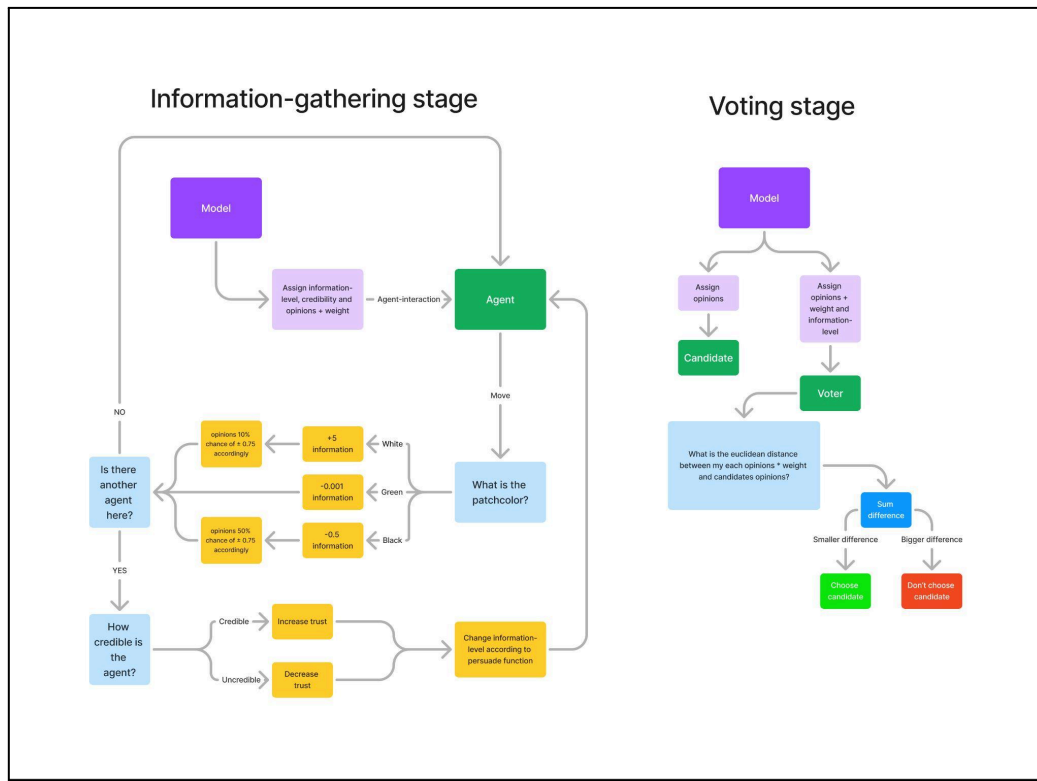


Figure 5.1: ABM workflow (For larger image, see appendix 3)

## 5.4 Data Analysis

To conduct the data analysis, all data of each agent in each simulation was extracted into a .csv file, whereafter the data was loaded into a new jupyter notebook, where the data analysis was conducted using python in VS code (Van Rossum & Drake, 2009; Visual Studio Code, 2024). To account for variance, 25 simulations, of approx. 10.000 ticks were run in ABM 1, while 10 simulations on each simulation in ABM 2. Further, all simulations in ABM 1 had 500 agents, while ABM 2 had 500 voters and 2 candidates and the information-level threshold was set to 0 for all simulations.

## 6. Results

As there are two outcomes for the vote, 0 and 1, there are no negative values. The outcome is therefore binary and a generalized linear model with binomial logistic regression was chosen. Visual exploration revealed a slight favor towards candidate 0, this being defined as the “good candidate” with opinions in the extreme good, in the democracy condition.

## 6.1 H.1 The effects of restricted suffrage

To test this hypothesis, agents with information-level less than 10 were excluded in an effort to simulate people not passing a hypothetical knowledge test to grant a vote. To investigate the effects of removing these agents, descriptive statistics were conducted for each condition as well as visualization.

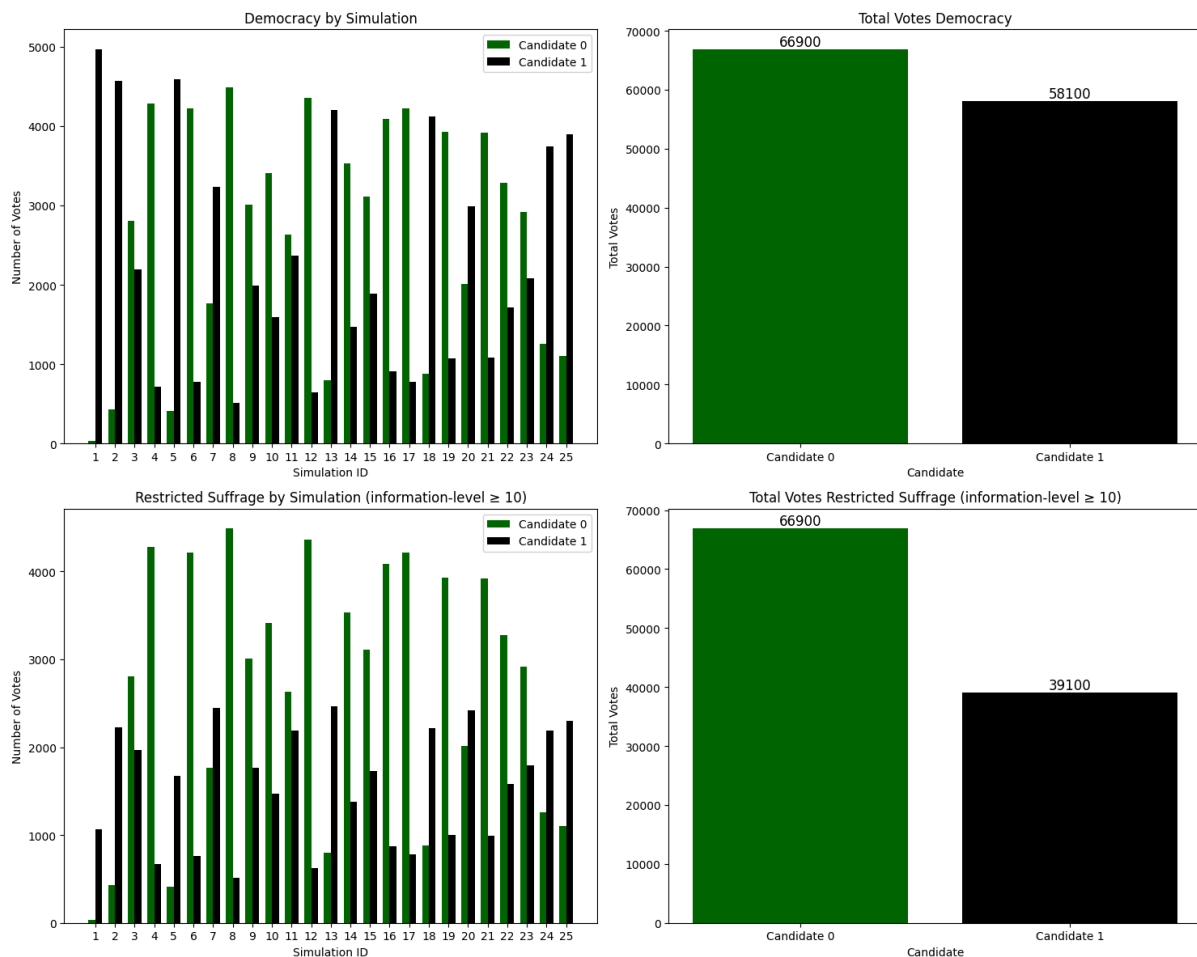


Figure: 6.1: Visualization of democracy vs restricted suffrage

These plots look quite different and skew the distribution of votes towards the good candidate, the means also differentiate: *Democracy*: 0.46; *Restricted suffrage*: 0.37. A Generalized Linear Model for a binomial logistic regression was also conducted. The model explained a moderate proportion of the variance in voting behavior, as indicated by a pseudo  $R^2$  value of 0.23. The analysis revealed a significant negative effect of the intercept: ( $\beta = -0.54$ ,  $SE = 0.006$ ,  $z = -84.37$ ,  $p < .001$ ), indicating that, in the absence of other predictors, there is a baseline preference for Candidate 0. However, excluding agents with information-level under 10 did not have a statistically significant effect on voting behavior: ( $\beta = 30.10$ ,  $SE = 11600.00$ ,  $z = 0.003$ ,  $p = .998$ ), suggesting that this did not substantially influence the likelihood of voting for candidate 0. This makes sense, as the

preference was already in favor of the “good” candidate, so removing some agents that would have voted for the “bad” candidate, doesn’t impact good governance. The only reason for this would be to make sure that the “good” candidate is chosen as they are rather close in the democracy condition.

## 6.2 H.2 Effects of plural voting

To test this hypothesis, agents with information-level above 50 are afforded an extra vote and agents with information-level above 75 are afforded one further extra vote, putting the total number of votes to 2 and 3 respectively, while agents with information-level under 50 still have 1 vote. To investigate the effects, the vote counts were visualized and descriptive statistics were conducted. A Generalized Linear Model for logistic regression was also conducted.

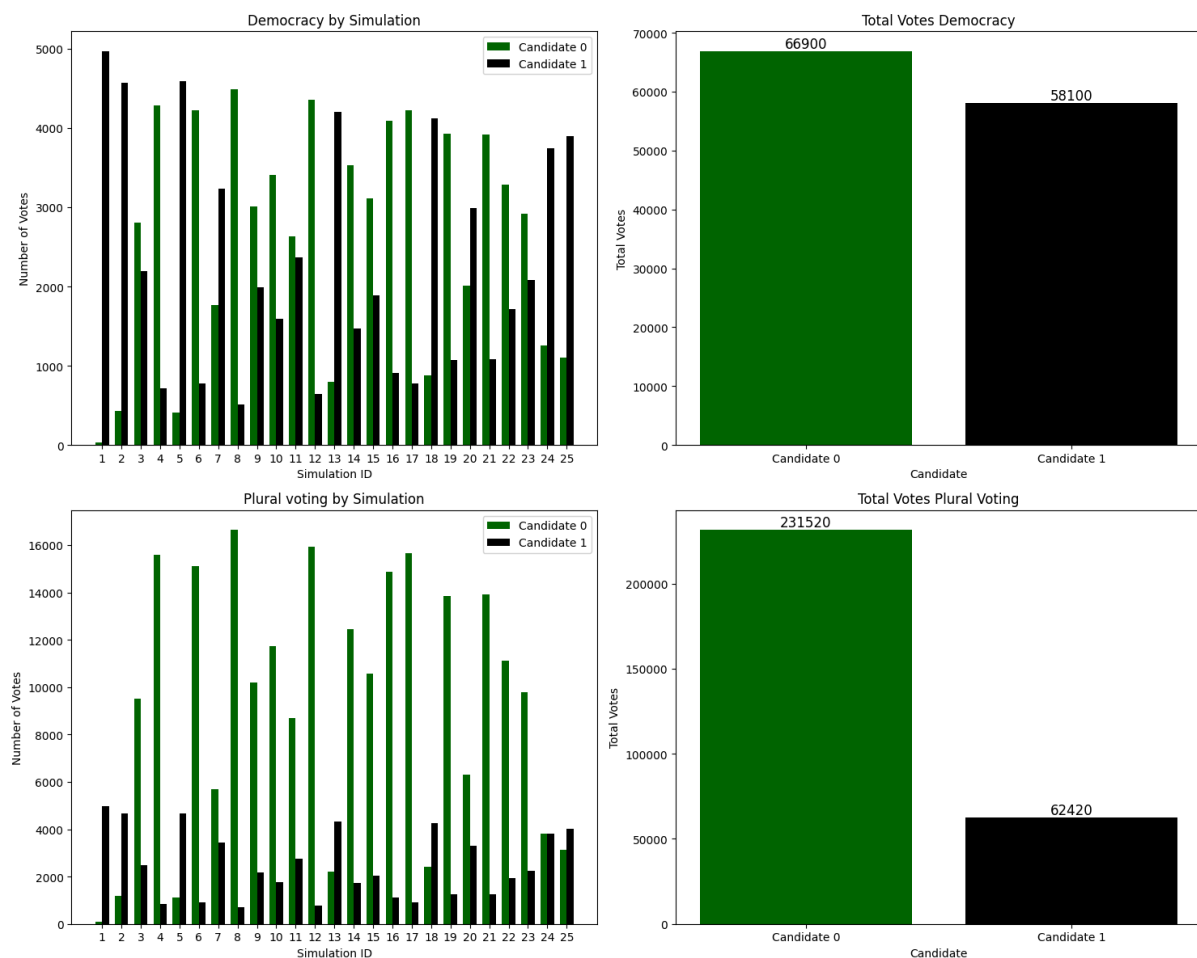


Figure 6.2: Visualization of democracy and plural voting

The means differentiate here as well: *Democracy*:0.46; *Plural Voting*: 0.21. Additionally, the visualizations are also noticeably different.

Again a Generalized Linear Model for a binomial logistic regression was conducted. This model explained a moderate proportion of the variance in voting behavior, as indicated by a pseudo  $R^2$  value

of 0.5420. The analysis revealed a non-significant effect of the intercept ( $\beta = 30.57$ ,  $SE = 11400.00$ ,  $z = 0.003$ ,  $p = .998$ ), indicating that, in the absence of other predictors, there is no meaningful baseline effect on voting for Candidate 0. Additionally, multiplying votes did not have a statistically significant effect on results ( $\beta = -33.85$ ,  $SE = 11400.00$ ,  $z = -0.003$ ,  $p = .998$ ), suggesting that multiplication did not substantially influence the likelihood of voting for Candidate 0.

## 7. Discussion

The present paper aimed to investigate the effects of changing the political system. This involved analysis of the mean vote and distribution of votes. To accomplish this, two ABMs were developed to simulate an environment in which agents interacted with each other and the landscape, to get information, whereafter they voted. The study found no significant difference in the different conditions. However, analysis of visual representations of all conditions revealed that changing the political system from democracy to epistocracy in either form will yield a more reliable outcome seeing as the difference is bigger.

### 7.1 ABM choices

The function of an ABM is to model reality rather than exactly replicating real world phenomena. Limitations, assumptions and complications will need to be carefully evaluated, when interpreting the results.

Limitations of the model include the simplicity. Information gathering and voting are complex phenomena that can be broken down in many categories and have many influencing factors. For these models it was a deliberate choice to simplify it, as factoring in everything would be too time consuming and out of scope. It is important, however, to acknowledge the complexity of political systems and the factors involved in determining how they function. These cannot be accurately captured in a model.

Another challenge was the quantification of information-level. People have varying levels of information and diverse interests. Moreover, information acquisition is not linear. Misinformation is a big factor in the real world and some information and knowledge might even be wrong. It was decided to make the information change more probabilistic, so not every piece of information was taken to heart, in an effort to simulate critical thinking. This will however not completely capture the global information landscape.

It is also worth noting the mechanisms for trust and credibility in ABM 1. The credibility of a real person can change a lot, whereas in the ABM it stays the same. Moreover, the trust characteristic, can,

if the ABM runs for too long, mean an inflated influence as it will increase over time in relation to meeting agents.

Lastly, as the simulations all used 500 agents, the effects are not necessarily generalisable over different populations and agent/landscape densities.

## 7.2 ABMs for social dynamics

It is important to reflect on the usefulness of ABM in modelling social dynamics. ABMs offer valuable insight into real complex phenomena, however should be noted that they also inherently involve simplifications and sometimes oversimplifications of these phenomena. Moreover, the possibility of constructing a model that superficially reflects real phenomena, without reflecting underlying dynamics, exacerbates the need for validation and causal power.

While the model reflects the expectations in the literature, it has limited use in terms of suggesting a change of political system.

## 7.3 Further research

For further research and analysis, the simulations could be conducted for different populations and densities. Additionally, more parameters for information-gathering could be included. Lastly, more simulations could be conducted to investigate the reproducibility and soundness of the results of this paper.

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## 9. Appendices

### 9.1 Appendix 1: Persuade function

```

to persuade
  ifelse ignorant [
    if any? other turtles-here with [under-informed or informed or well-informed] [
      if random-float 1 < 0.25 [
        set information-level information-level + (0.001 * trust)
      ]
    ]
  ] [
    ifelse under-informed [
      if any? other turtles-here with [informed or well-informed] [
        if random-float 1 < 0.1 [
          set information-level information-level + (0.002 * trust)
        ]
      ]
    ] [
      ifelse informed [
        if any? other turtles-here with [well-informed] [
          if random-float 1 < 0.1 [
            set information-level information-level + (0.0025 * trust)
          ]
        ]
      ] [
        if any? other turtles-here with [well-informed] [
          if random-float 1 < 0.01 [
            set information-level information-level + (0.001 * trust)
          ]
        ]
      ]
    ]
  ]
] [
  ifelse well-informed [
    if any? other turtles-here with [ignorant or under-informed or informed] [
      if random-float 1 < 0.005 [
        set information-level information-level - (0.001 * trust)
      ]
    ]
  ] [
    ifelse informed [
      if any? other turtles-here with [ignorant or under-informed] [
        if random-float 1 < 0.25 [
          set information-level information-level - (0.002 * trust)
        ]
      ]
    ] [
      ifelse under-informed [
        if any? other turtles-here with [ignorant] [
          if random-float 1 < 0.1 [
            set information-level information-level - (0.01 * trust)
          ]
        ]
      ] [
        if any? other turtles-here with [ignorant] [
          if random-float 1 < 0.5 [
            set information-level information-level - (0.01 * trust)
          ]
        ]
      ]
    ]
  ]
]
end

```



## 9.2 Appendix 2: Evaluate\_candidate() function

```
def evaluate_candidate(self):
    min_distance = float('inf')
    chosen_candidate = None

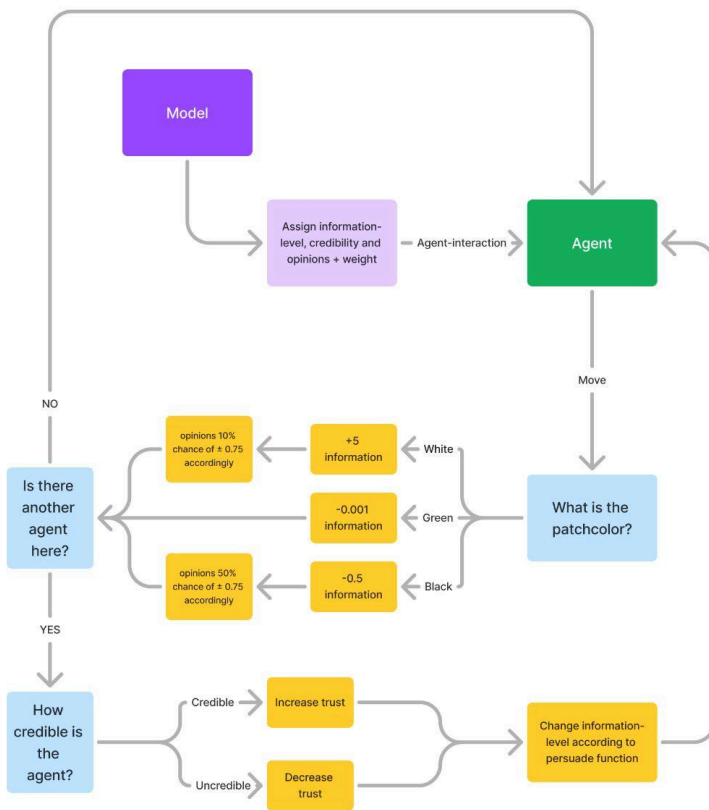
    for candidate in self.model.candidates:
        # Calculate the weighted Euclidean distance
        distance = sum(
            self.attributes[f"{attr}-weight"] * ((self.attributes[attr] - getattr(candidate, attr)) ** 2)
            for attr in [
                "government-intervention",
                "government-control-of-economy",
                "free-trade",
                "protectionism",
                "pro-choice",
                "tax-increase-to-offset-debt-and-deficit",
                "crime-punishment-level",
                "military-hawkishness",
                "non-military-interventions",
                "affirmative-action",
                "prayer-in-schools",
                "market-solution-to-healthcare",
                "moralistic-law",
            ]
            if attr in self.attributes and f"{attr}-weight" in self.attributes
        )
        # Take the square root of the weighted squared differences
        distance = distance ** 0.5

        if distance < min_distance:
            min_distance = distance
            chosen_candidate = candidate

    return chosen_candidate.unique_id # Return the chosen candidate's unique ID
```

### 9.3 Appendix 3: Workflow of ABMs full size

#### Information-gathering stage



#### Voting stage

